KuKu — Find the Promos: System Submission

Track: **Find the Promos** · Default mode: **Hybrid** · Outputs: **By-episode catalogue** + **Campaign Priority pack** · Built with **Grok** (xAI) as the main coding partner

1. PROBLEM I PICKED — AND WHY

I chose **Find the Promos**. Kuku's shows win attention in the feed: a short cut has seconds to stop a thumb, tell a tiny complete story, and leave an open loop. Hand-picking across ~50 episodes is slow, uneven, and biased toward whatever the editor watched last. The job is not "find spicy lines" — it is **find shippable storytelling windows**, then decide **what the whole series should post first**.

Side note — human psychology behind the approach. *When I watch a short or promo for a show or web series, I rarely care if that moment is from the pilot or episode 30. If it feels good — clear open, rising pressure, unfinished ending — I jump to the show. Cold audiences buy a feeling and a question, not a plot lecture. So we built for* ***standalone mini-stories that still sound like this series****, not first-scene-only trailers and not raw dialogue dumps.*

2. ASSUMPTIONS (INCOMPLETE INFO → KEEP MOVING)

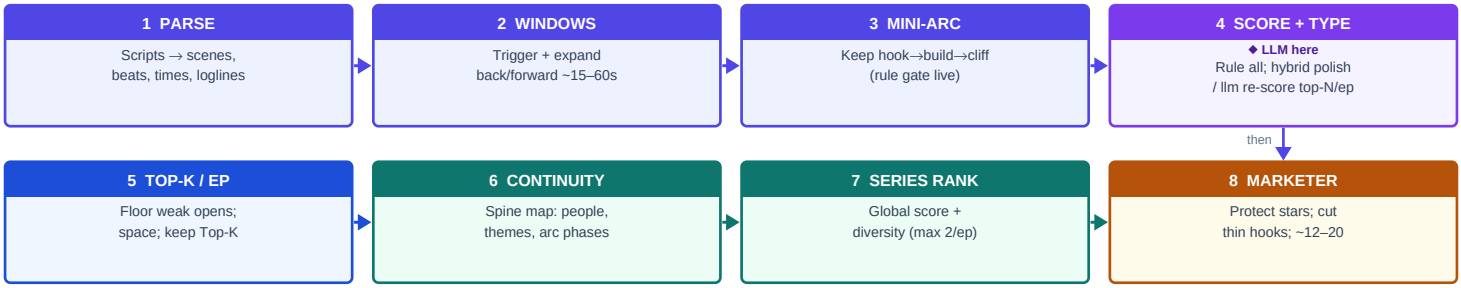
The brief leaves ops gaps. I treated that as the job: make a sensible call, write it down, ship.

- **Scripts are source of truth** for structure/timing (video is downstream).
- **Most viewers are cold** — standalone clarity ranks with drama heat.
- **Target ~15–60s** storytelling windows (pack QA ~14–62s).
- **Two questions, two lists:** best-in-episode ≠ post-first for the show.
- **Labels are tags, not gates** — strong soft beats can still win.
- **API keys/quotas can vanish** — default must run on rules and degrade honestly.
- **One project = one show**; **humans stay final** on ship decisions.

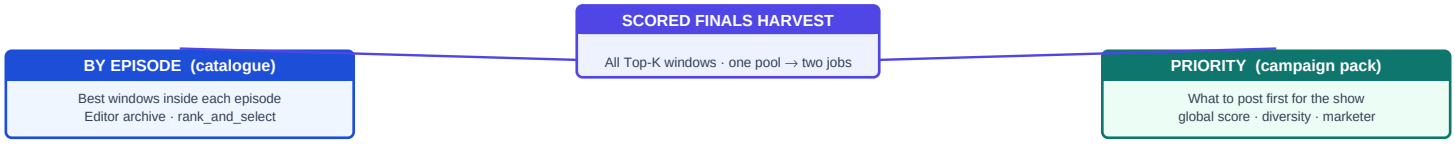
3. HOW THE SOLUTION WORKS

Studio line we mirrored: editor marks every episode → pool fills → series memory informs taste → growth ranks the feed → marketer protects stars / kills weak cuts → board reviews.

PIPELINE · how a show becomes a promo pack



1 · Parse. Scripts → scenes/beats with characters, dialogue, timestamps — timed ranges, not loose quotes.
2 · Windows (always rule). Find triggers, expand **backward and forward** into ~15–60s storytelling spans (sub-windows inside long scenes). Mode never shrinks search — coverage stays complete and cheap.
3 · Mini-arc gate. Keep only windows that play **hook** → **escalation** → **cliff**. Live runs use the fast rule validator; LLM mini-arc code is ready when paid capacity is wired (page 2).
4 · Score + type ◆ LLM (optional). Rule baseline for everyone (0–10 hook/cliff/emotion/narrative/overall + types). **Hybrid default:** keep rule numbers; LLM polishes top-N/ep (reason, hook_type, types) for human review. **LLM mode:** model may re-score that top-N. No key/errors → pure rule. Long tail stays rule — we do not LLM-score every weak window.
5 · Top-K per episode. Floor weak opens/ends; prefer spaced moments; keep Top-K (often 5). Feeds **By episode** and the campaign harvest pool.
6 · Continuity map. Derived from this show's episodes: people, spine themes, arc phases, spoiler guidance — living memory, not a one-title hand bible.
7 · Series rank + diversity. Each final gets **series_global_score** (quality + hook/cliff + continuity fit + clarity + role + duration − spoiler; **types excluded**). Diversity pack (max 2/ep, soft type/theme spacing) → ~20–30 shortlist.
8 · Marketer (rule in web path). On enriched hook/cliff text: protect top/high-global rows; hard-cut thin hooks and broken segments; soft multi-fail drops; size-trim ~12–20 → **Priority**.



Global score weights (code): 0.26 overall · 0.16 hook · 0.16 cliff · 0.08 emotion · 0.06 narrative · 0.14 continuity fit · 0.10 clarity · 0.04 role · 0.04 duration · –0.06 spoiler (+ small acquisition/fit bonuses). Generate writes artifacts; reopening a project reloads the last save — no silent full re-score.

KuKu — Find the Promos: System Submission (page 2)

How we got here · modes & paid LLM path · scaling · proof · AI use · next improvements

4. HOW THE DESIGN EVOLVED (INCOMPLETE INFO → DECISIONS)

Early AI drafts leaned “smart agents everywhere.” Elegant on a whiteboard; painful at season volume. Path we walked:

Stage	Tried	Learned → decision
Agent-per-step	Sequential LLM: expand→validate→score→classify→rank	Latency/cost explode. Structure stays rule-first; no agent chain on every beat.
Score-only rules	Pure heuristics end-to-end	Great coverage/uptime; cards can feel flat → optional LLM polish.
Full LLM scoring	Model judges everything	High ceiling, brittle on keys/quotas; overkill for long-tail windows.
Hybrid (landed)	Rule core + LLM top-N/ep polish (or re-score in LLM mode)	Default: always ships, bounded spend, better review cards when API is up.
Types as filters	Kill “wrong” genres early	Burials strong oddballs. Types = tags; formula ignores them; soft pack diversity only.
One ranked list	Single scoreboard for UI	Confuses archive vs feed → By-episode catalogue + Priority campaign pack.

5. RULE · HYBRID · LLM — PAID CAPACITY RAISES THE CEILING

Mode	Scores	LLM role	When
Rule	Heuristics all	—	Bulk / offline / zero spend
Hybrid ★	Stay rule	Polish top-N/ep (reason, types, hook_type)	Default product path + auto fallback
LLM	May re-score top-N/ep	Full score pass on shortlist	Paid capacity + launch taste pass

Why Hybrid is default. A promo tool that dies without a key is not a product. Rule floor on every window; tokens only where humans read (top few/ep); clean fallback if Groq is missing or rate-limited. Full eps 1–50 head-to-head: **[LINK: MODE_REPORT — output/full_compare/MODE_REPORT.md]**.

Paid LLM path (turn the dial, don’t rewrite). Scoring uses Groq when `GROQ_API_KEY` exists. Free/low tiers fine for Hybrid polish; full-season LLM mode hits RPM/TPM and daily caps (we already saw quota-shaped degradation). With paid capacity:

- **Groq paid / higher limits (Performance or Batch-style):** raise concurrency + `llm_per_episode`; true LLM re-score on a larger shortlist; wire existing mini-arc LLM/hybrid for borderline structure. Fast open-weight models keep 50-ep harvest cost sane.
- **Batch / Flex-style (Groq Batch, or OpenAI Batch ~half realtime):** overnight full-catalogue taste passes — score + mini-arc offline, packs ready by morning. Best for every-show weekly ops.
- **Frontier model (OpenAI/Anthropic) on final Priority only (~20–30):** brand voice, spoiler nuance, copy quality after rule harvest + global rank — never on thousands of raw windows.
- **Still rule-first:** window expansion + long tail. Paid LLM multiplies shortlist judgment; it must not revive sequential agents on every beat.

6. SCALING TO EVERY SHOW WE MAKE

Shape fixed; content per-title. **Projects** isolate scripts/outputs (create / generate / delete) — one app, many shows. Structure-first intake (scenes/beats) plugs new packs without a stack rewrite. Continuity is **derived from that show’s episodes**. Campaign role scales with season length (early acquisition / late payoff as soft prior). Mode dial: rule bulk · hybrid daily · llm when paid capacity + launch week. By-episode + Priority stay the two operator surfaces. Per-show care is intentional: light theme lexicon + gold slices as genre mix grows — localization of memory/eval, not a pipeline fork.

7. HOW WE KNOW IT’S WORKING

Each promo is a window with hook, cliff, rationale, scores, campaign context — argument-ready, not a keyword dump. Dual lists are real separate paths (episode Top-K vs full-finals pack + marketer). Mode compare on eps 1–50 covers agreement, unique finds, quota degradation (**[LINK: MODE_REPORT]**). `evaluation/` supports gold-style checks beyond UI vibes. Web Generate runs the real orchestrator + deliverable writers, so UI surfaces match emitted objects. Bar: a cold teammate opens Priority, gets why #1 is first, checks By episode coverage, glances Continuity, exports — no tribal sheet.

8. HOW I USED AI — AND WHERE IT GOT IT WRONG

Primary partner: **Grok (xAI)** for implementation, refactors, web UI, mode plumbing, marketer/ranker iteration, and dense docs — always re-checked against `src/`.

- **Sequential agents everywhere.** Early Grok lean: LLM expand→validate→score→classify→rank. That would crawl on full seasons. I forced rule structure + bounded LLM only on top-N polish.
- **Score-only or pure-LLM defaults.** AI collapsed “good promo” into one scoreboard or full model taste. We tried both; neither is product default. Hybrid won for uptime + honest fallback + peak polish.
- **Types as hard filters.** Demo-convenient, growth-wrong. I kept types as tags; global formula excludes them.
- **Over-claiming docs.** Drafts said hybrid “re-scored judgment” and mini-arc LLM was live. Code audit fixed it: hybrid polishes copy/types; live mini-arc is rule-wired. Verify against code, not vibes.
- **UI scope creep.** Asked for one control (e.g. project delete); agent rewrote unrelated surfaces. I enforced narrow diffs only.

Pattern: AI accelerates scaffolding; I own architecture, constraints, and “does the code match the sentence?”

9. WHAT’S SOLID · NATURAL NEXT IMPROVEMENTS

Solid today: storytelling windows end-to-end, dual catalogue/campaign layers, continuity-aware global rank with explicit weights, marketer protect/reject, projectized web app, Hybrid default with true rule fallback, mode-compare on full 1–50 harvest. **Next investments (same architecture, higher ceiling):** wire mini-arc LLM on paid capacity for borderline saves · batch overnight catalogue taste passes · frontier pass only on final Priority shortlist · per-genre continuity lexicons + gold sets · tighter timing robustness and optional handoff into edit tools.

10. POINTERS

Artifact	Where
This 2-page submission	docs/KuKu_Promo_Finder_Submission.pdf
Code-verified overview	docs/SYSTEM_OVERVIEW.md
Mode comparison (eps 1–50)	[LINK: MODE_REPORT — output/full_compare/MODE_REPORT.md]
Decision log	DECISIONS.md
Pipeline core	src/orchestrator · candidate_generator · mini_arc_validator · llm_agents · series_ranker · marketer_agent · final_deliverable
Web + eval	web/app.py · web/services/data_layer.py · web/static/ · evaluation/

Bottom line: promos are mini-stories; catalogue ≠ campaign; Hybrid is the honest default; paid LLM raises shortlist ceiling without slow agent chains. Grok sped the build — human judgment set constraints and caught over-claims.

KuKu AI Generalist Assignment · Find the Promos · 2-page submission · diagrams verified against live pipeline code